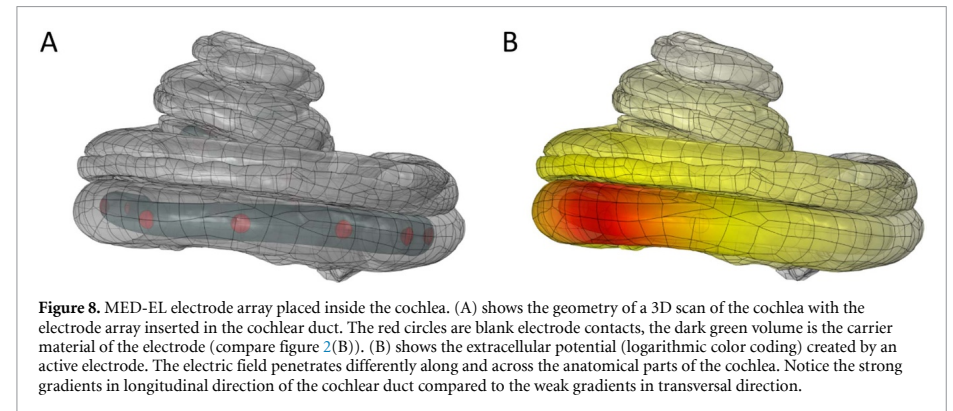
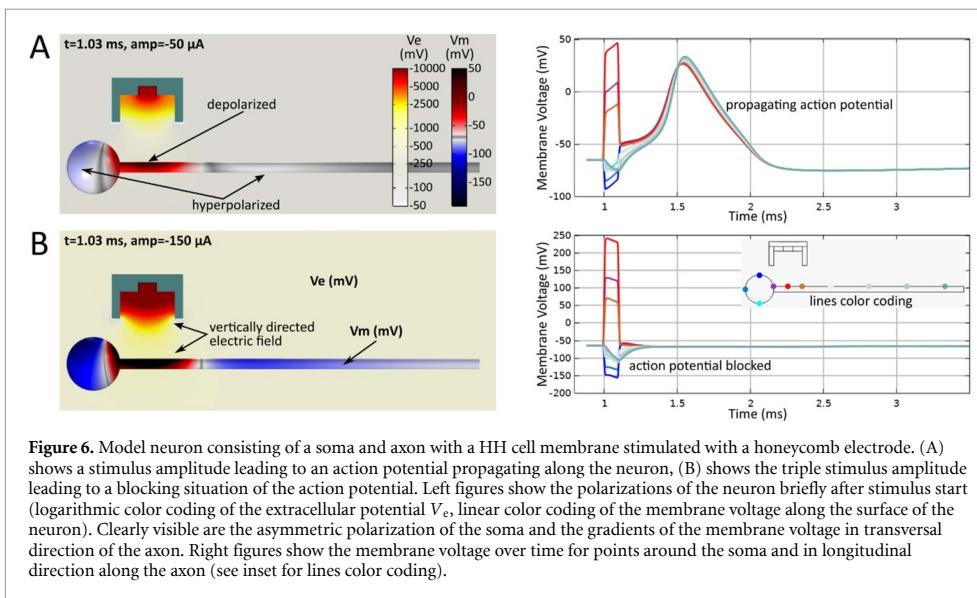


Finite element modelling (FEM) for extracellular neural stimulation

Supplementary slides inspired by Fellner et al., Journal of Neural Engineering, 2022



why model?

current practice

what FEM adds

examples

references

Summary

1. Motivation

Why computer models are needed in extracellular neural stimulation.

2. State of the art

Analytical, hybrid and whole-FEM approaches.

3. FEM potential

Added values of finite element modeling when the geometry is complex.

4. Workflow

Main modelling steps: geometry, mesh, physics, solver, results.

5. Examples

Honeycomb electrodes, cochlear implant and myelinated fibre cases.

6. Limits

Computational cost, model complexity and when FEM is justified.

Why modelling matters

In stimulation, we usually cannot directly see the electric field, the electrical potentials or the membrane response.

Interpret experiments

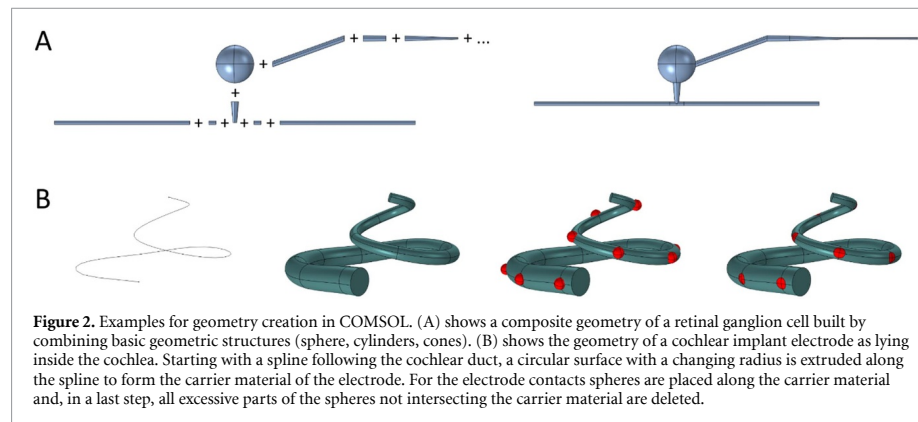
Estimate which neural structures are activated, where the action potential starts, thresholds values, etc.

Design devices

Compare electrode shapes, materials, stimulus waveforms and locations before building prototypes.

Predict therapy effects

Link the stimulus to targeting, selectivity, side effects and patient-specific anatomy.



Key idea: when the electrode, the tissue and the neuron influence each other, the model becomes part of the experiment.

What is commonly done now?

Most models still combine simplifications with efficient numerical tools.

Analytical field models

Point-source or disc-electrode approximations. Useful for insight but assume simple electrode/tissue conditions.

Multi-compartment cells

The neuron is split into cylindrical or conical compartments; membrane dynamics are solved efficiently.

Hybrid approach

FEM computes the electric field; NEURON, MATLAB or another solver computes the neural response.

Typical benefits

- fast simulations
- well-established tools
- good for many standard questions

Typical limitations

- simplified geometry
- weak interaction between cell and extracellular space

What FEM adds

Finite elements solve the physics on the actual geometry.

Realistic 3D geometry

Electrodes, carrier material, neuron morphology, ducts, tissue boundaries and implants.

Distinct materials

Different conductivity and permittivity values; isotropic or anisotropic tissues.

Boundary conditions

Current or voltage terminals, grounds, insulating boundaries and active membrane fluxes.

Spatial gradients

The membrane can experience local, asymmetric and transverse potential gradients.

Coupled physics

Transmembrane current affects the extracellular field and the field affects membrane voltage.

Multiphysics extensions

Electrode impedance, tissue capacitance, heating and ionic diffusion can be added.

FEM is most useful when the geometry is not a detail but the main reason why stimulation works or fails.

The whole-FEM concept

Cell, extracellular space and membrane equations are solved in one coupled model.

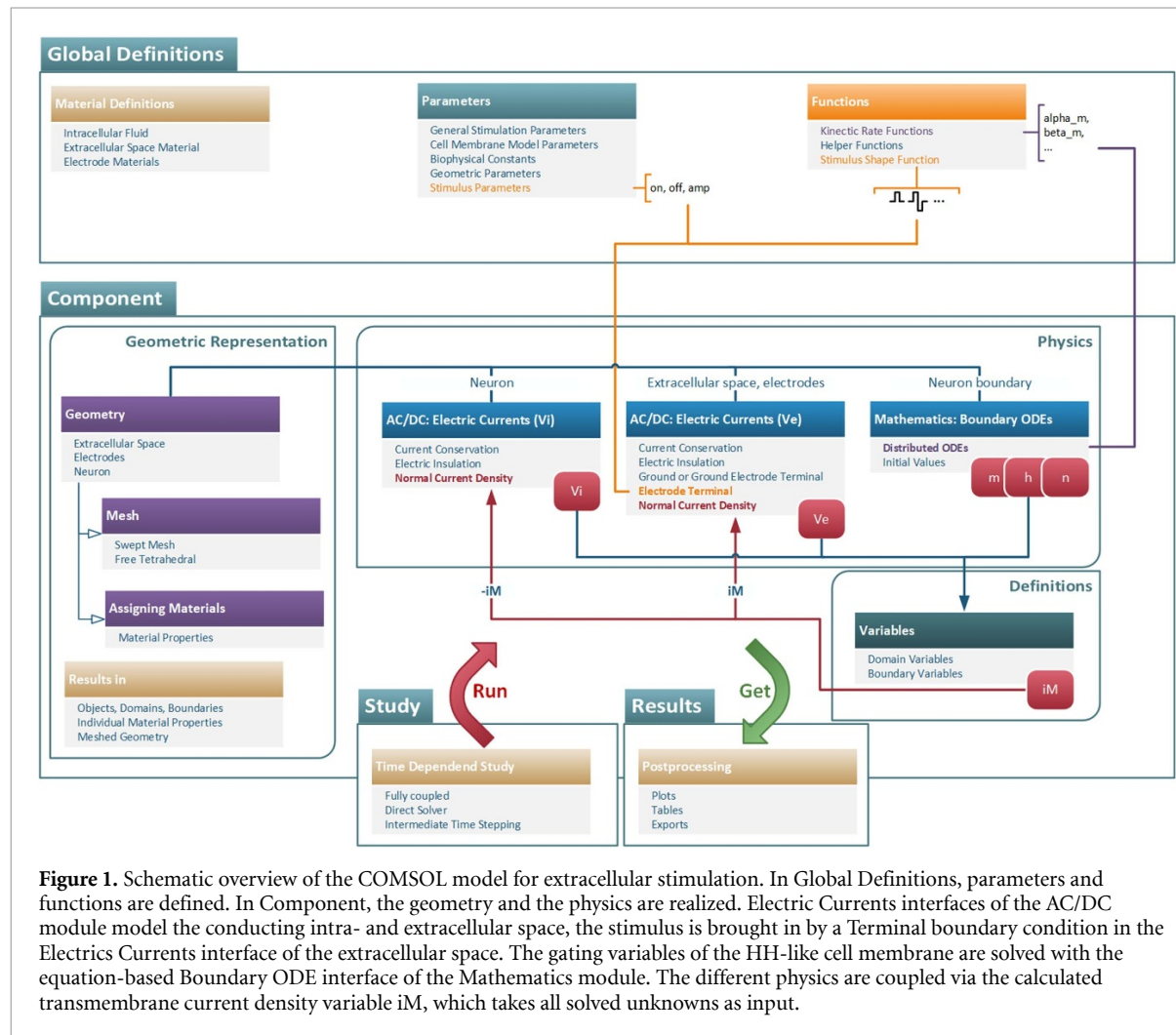


Figure 1. Schematic overview of the COMSOL model for extracellular stimulation. In Global Definitions, parameters and functions are defined. In Component, the geometry and the physics are realized. Electric Currents interfaces of the AC/DC module model the conducting intra- and extracellular space, the stimulus is brought in by a Terminal boundary condition in the Electric Currents interface of the extracellular space. The gating variables of the HH-like cell membrane are solved with the equation-based Boundary ODE interface of the Mathematics module. The different physics are coupled via the calculated transmembrane current density variable iM , which takes all solved unknowns as input.

Extracellular space

Electric Currents interface solves for extracellular potential V_e .

Intracellular space

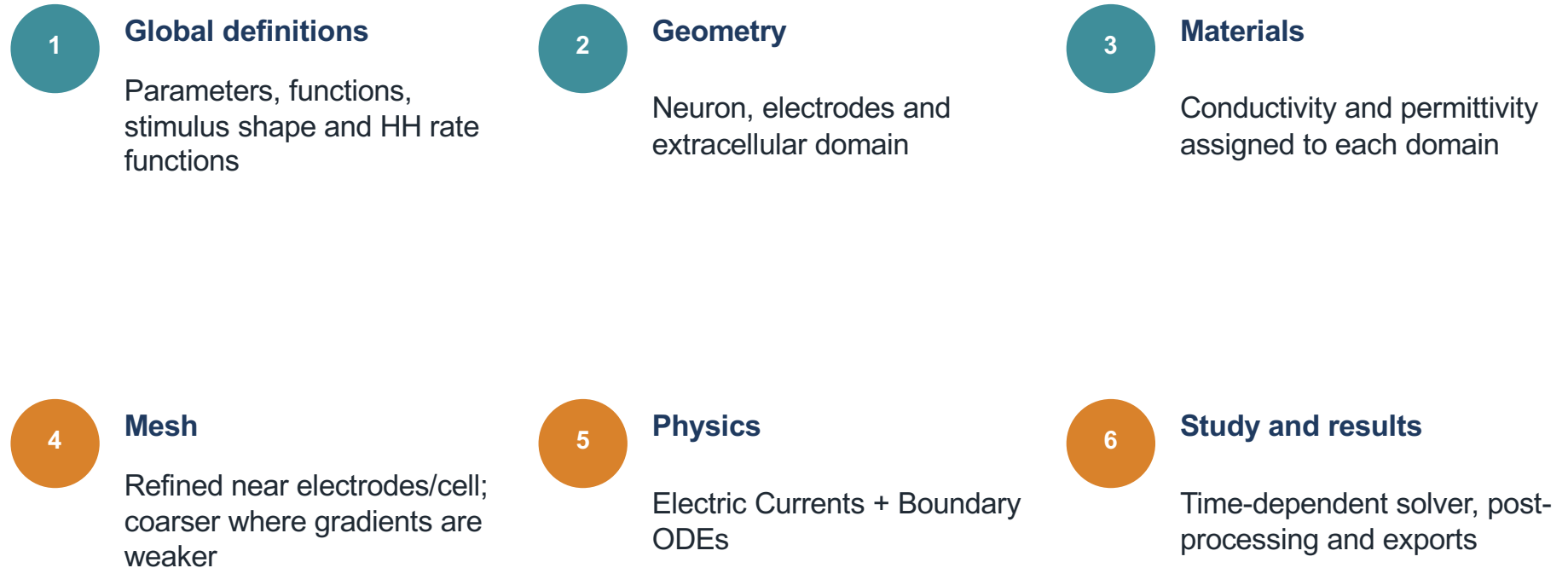
A second Electric Currents interface solves for intracellular potential V_i /

Cell membrane

Boundary ODEs solve gating variables; current density iM couples both spaces.

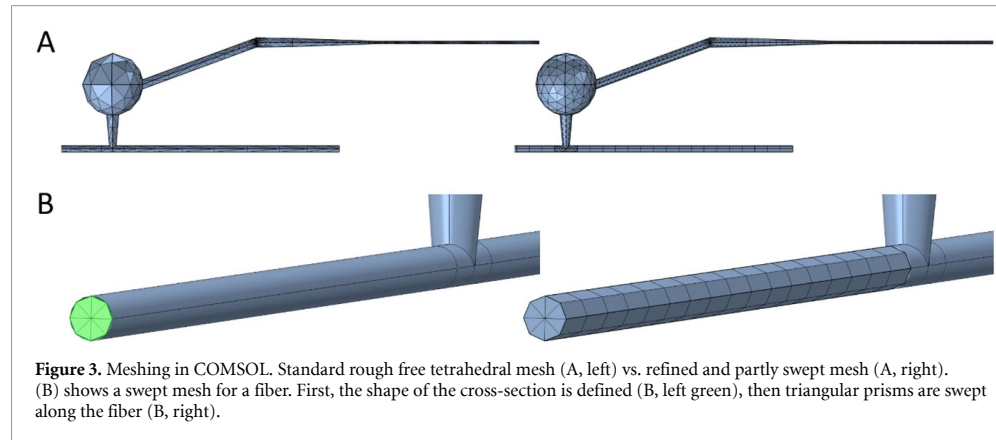
All unknowns are solved together at each time step!

Implementation workflow in COMSOL



Geometry and meshing choices are modelling choices

The mesh controls accuracy, degrees of freedom and runtime.



Refine where needed

Near electrodes, membranes and regions with strong field gradients.

Simplify where possible

Far from electrodes, a lower spatial resolution may be sufficient.

Use swept meshes

Fibres can be meshed along their length with controlled cross-sections.

A good FEM model is not the most detailed model. It is the simplest model that can answer the question with enough accuracy.

Membrane model inside FEM

The Hodgkin-Huxley formalism is implemented on the cell boundary.

Membrane voltage

$V_m = V_i - V_e$, evaluated locally on the membrane boundary.

Capacitive current

Depends on membrane capacitance and the time derivative of V_m .

Ionic current

Uses HH-like gating variables and reversal potentials.

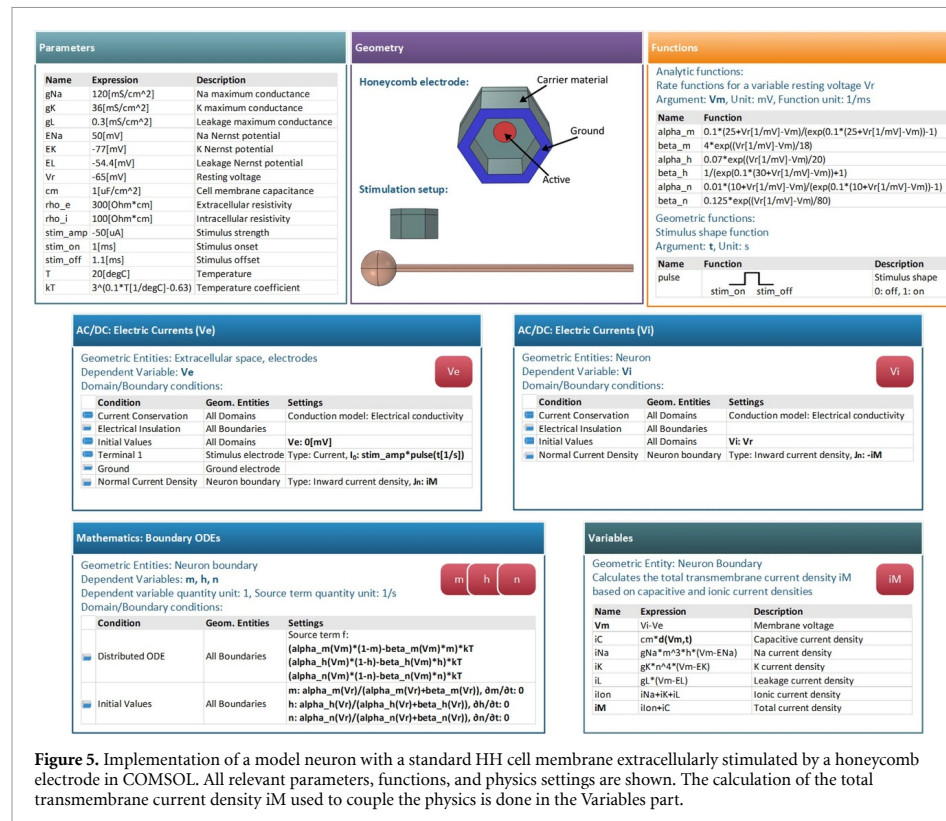
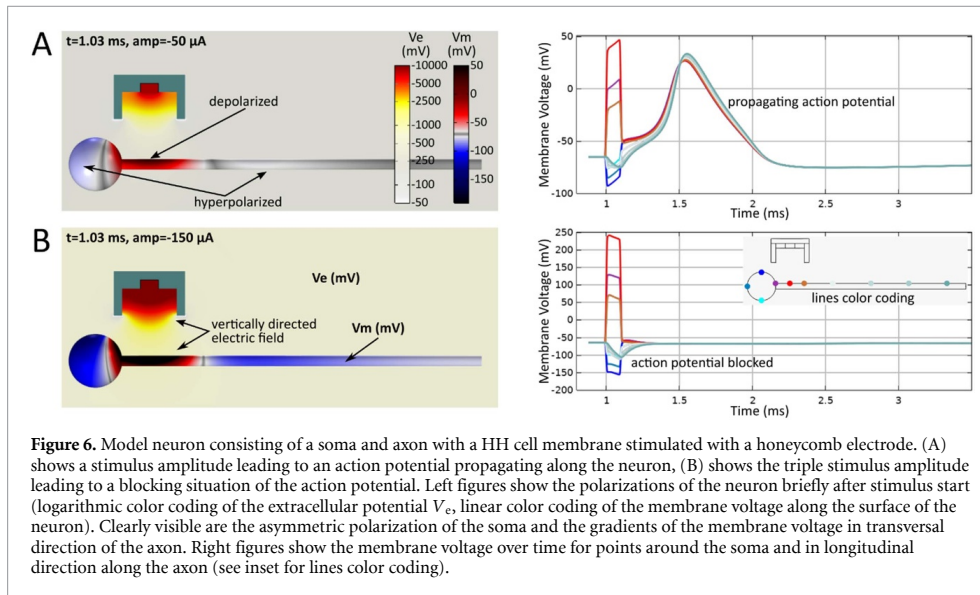


Figure 5. Implementation of a model neuron with a standard HH cell membrane extracellularly stimulated by a honeycomb electrode in COMSOL. All relevant parameters, functions, and physics settings are shown. The calculation of the total transmembrane current density iM used to couple the physics is done in the Variables part.

Examples and ideas for projects

Example 1: honeycomb electrode + HH neuron



Setup

Model neuron with soma and axon; HH membrane; honeycomb electrode.

Moderate stimulus

Generates a propagating action potential.

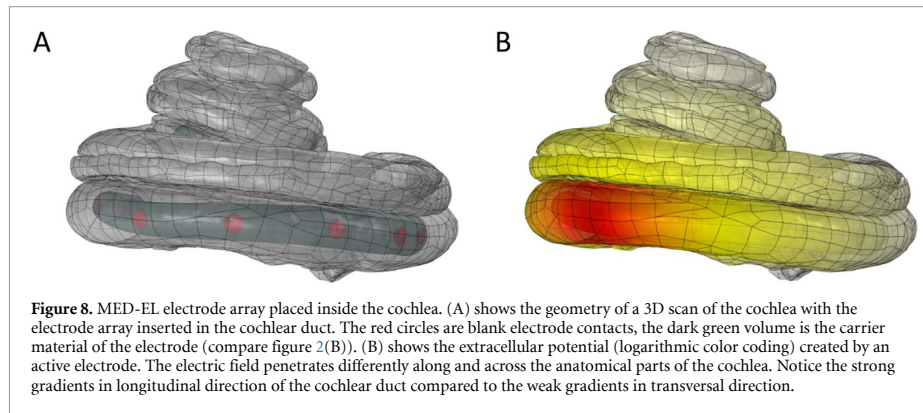
Strong stimulus

Can block propagation

Take on message: stimulation amplitude, electrode geometry and cell morphology interact; more current is not always equivalent to more activation.

Example 2: cochlear implant electrode

Here the main challenge is the anatomy and material heterogeneity.



Complex geometry

The electrode array follows the cochlear duct; contacts and carrier material are represented.

Different tissues

Duct fluids and bone-like structures have different electrical properties.

Model output

The extracellular potential distribution can be exported to fibre models.

A realistic field model can be valuable even if the neural membrane is then simulated with a faster method.

Example 3: myelinated fibre and MRG model

A detailed double-cable model can also be represented in FEM.

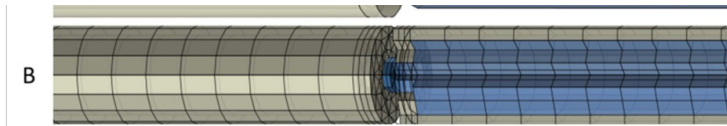


Figure 9. Geometry and swept mesh of the myelinated fiber. (A) shows the myelinated internode (left), the Node of Ranvier (center, dark blue), and the axon without a surrounding myelin sheet (right). (B) shows an example swept mesh for the myelinated fiber (left) and the cross-section of the swept meshed fiber (right).

myelinated nerve fibers, which is commonly used for them accordingly. A swept mesh consisting of tr

Problem

The periaxonal space is only a few nanometres thick and cannot be meshed as a normal 3D volume.

Solution

Use Electric Currents in Layered Shells to model it as a conducting boundary layer.

Result

Simulate AP propagation and quantify how the fibre activity changes local V_e .

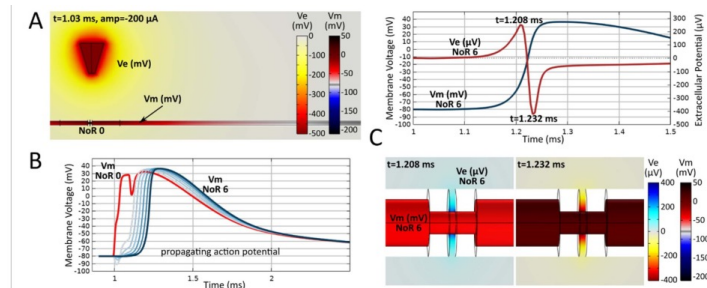


Figure 10. Extracellular stimulation of a myelinated nerve fiber with a MRG cell membrane. (A) shows the electric field of an active conical electrode placed above a node of Ranvier (NoR) briefly after stimulus start. (B) shows the membrane voltage V_m at several adjacent nodes of Ranvier. An action potential is first triggered at NoR 0 below the electrode (red) and propagates along the fiber from NoR 1 (light blue) to NoR 6 (dark blue). (C) shows the effects on the extracellular potential V_e when the action potential passes NoR 6. During this time no stimulus is active, V_e is only influenced by transmembrane currents. In (C, top), V_e (red) in the nodal gap of NoR 6 first becomes positive when the membrane voltage (blue) increases, which is caused by capacitive currents. Later, when ionic currents due to the highly active cell membrane set in, V_e becomes negative, before returning back to rest. In (C, bottom), the nodal gaps are shown at the respective peaks of V_e , left at $t = 1.208$ ms when V_e is positive, right at $t = 1.232$ ms when V_e is negative.

Overview of all examples in the paper

Each example answers a different modelling need.

Example	Main question	Why FEM is useful	Main output
Honeycomb electrode + HH cell	Does the stimulus trigger or block an AP?	Directed field and 3D cell polarization	Vm maps, AP propagation, thresholds
Whole FEM vs hybrid	How much error comes from simplification?	Reference solution with full geometry	Lower and upper threshold comparison
Cochlear implant	How does the field penetrate the cochlea? Study the activation function	Anatomy, fluids, bone and electrode carrier	3D Ve distribution for auditory fibres
Myelinated MRG fibre	Can a detailed double-cable model be built?	Layered shell and boundary coupling	AP propagation and Ve generated by the fibre
Meshing/solver strategy	How can the model stay computable?	Control DOF without losing key gradients	Refined/swept mesh and time stepping

When should we use whole FEM?

Strong local gradients

Cells are close to electrodes or electrode design creates a directed field.

Complex extracellular space

The tissue has relevant geometry, material interfaces or anisotropy.

Cell-environment feedback

The neuron activity itself changes the field or affects nearby cells.

Multiphysics questions

Temperature, diffusion, electrode impedance or capacitive tissue effects matter.

Reference solution

A slower but more realistic model is needed to validate faster approximations.

Not always needed

If the field is simple and geometry is not critical, simpler models may be better.

Limitations and trade-offs

Computational cost

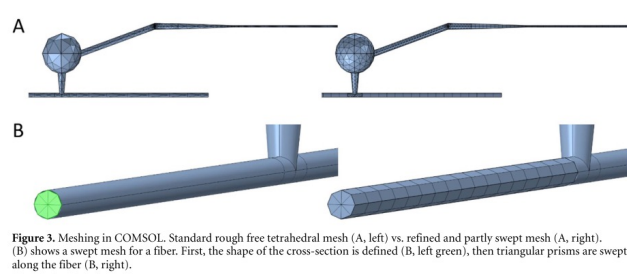
High spatial and temporal resolution can lead to long runtimes and large memory demand.

Model development effort

Building and validating geometry, mesh, coupling and solvers requires expertise.

Software cost

Professional FEM software such as COMSOL may require expensive licences.



Practical strategy

- begin with a simple model
- identify which simplification may bias the answer
- increase FEM detail only where it changes the result
- use whole FEM as a reference when needed

Take-home messages

1. Modelling is a bridge

It connects physics, device design and biological response.

2. Simplifications are choices

They are not wrong, but they must match the question.

3. FEM adds geometry

This is crucial when the field is shaped by electrodes, tissues or cells.

4. Whole FEM couples physics

The membrane and extracellular field can interact in space and time.

5. Examples guide use

Honeycomb, cochlea and MRG fibre show different use cases.

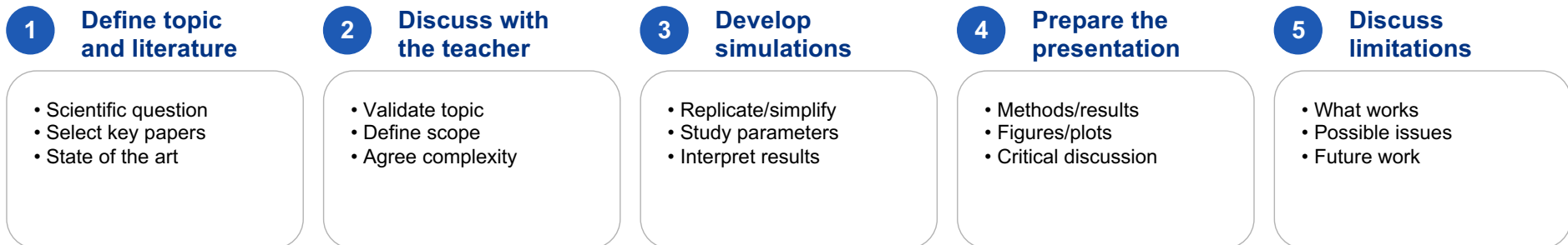
6. Cost remains important

Use FEM when the expected gain in accuracy justifies the effort.

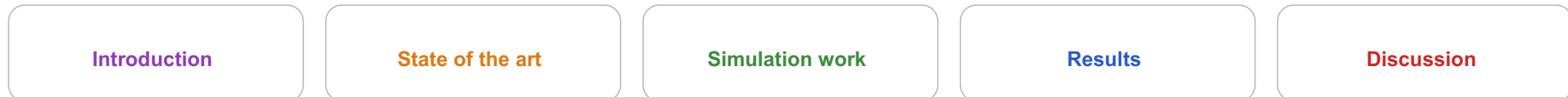
Some ideas for the project

- Honeycomb electrode + HH neuron ([Fellner 2022](#); literature search)
- Cochlear implant ([Fellner 2022](#); [Cochlear stim](#); literature search)
- Myelinated fibre ([Fellner 2022](#); literature search)
- Retinal models and stimulation ([Fellner 2022](#); [Dokos 2005](#); [ABUELNASR](#); literature search)
 - Abuelnasr is very complex
- Simulating epileptic seizures ([Schreiner 2022](#); literature search)
- Why simple models can fail? ([Fellner 2022](#); literature search)
- Deep brain stimulation models ([Butson 2006](#); [Butson 2007](#); literature search)
- Model electrode–tissue interface in electrical stimulation ([Merril 2005](#); literature search)
 - Complex topic!
- Other relevant application you may find in literature (see the bibliography for additional ideas).

Projects organization



THE FINAL PRESENTATION SHOULD INCLUDE



- Agree for a date of discussion
- Project should be delivered three days in advance
 - Exam will be registered at the official date
- NB: Please register for the nearest exam session!

Deliver both presentation and simulation codes (Matlab, Comsol)

Selected references

Core references cited in the supplementary slides.

- Fellner A, Heshmat A, Werginz P, Rattay F. A finite element method framework to model extracellular neural stimulation. *J Neural Eng.* 2022;19:022001.
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